

# LEARNING DUAL MIXTURE-OF-EXPERTS MODELS FOR UNIFIED IMAGE DERAINING

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## ABSTRACT

Despite the remarkable progress achieved in image deraining, most existing approaches remain constrained to handling a single type of rain degradation or a specific rain pattern. In practice, however, rain manifests in diverse forms, which severely limits the applicability of current methods under real-world conditions. To overcome this limitation, we first construct a high-quality benchmark named Hybrid-Rain10K, which integrates three representative rain degradation patterns, including rain streaks, raindrops and rain accumulation. Building upon this benchmark, we propose a dual mixture-of-experts (MoE) model for unified image deraining, which adaptively handles various rain degradations and restores clean images. Where the encoder employs a Soft-MoE to assign collaborative weights across all experts, ensuring comprehensive degradation modeling, while the decoder adopts a Hard-MoE with top-k routing guided by high-frequency priors to enhance reconstruction quality. Extensive experiments demonstrate the effectiveness of our dataset and model.

**Index Terms**— Unified image deraining, Mixture-of-Experts, Deraining datasets, Rain removal

## 1. INTRODUCTION

Rainy weather substantially degrades outdoor image quality, impairing both visual perception and the reliability of downstream vision tasks such as detection, tracking, and autonomous driving. Consequently, single image deraining has long been recognized as a fundamental yet challenging problem in low-level vision [1–3]. In recent years, numerous rain datasets [4–7] and diverse methods based on CNNs and Transformers have been proposed, incorporating priors and attention mechanisms to learn complex degradations and achieving promising results across benchmarks.

Nevertheless, real-world rain appears in diverse forms, including sparse or dense streaks, complex raindrop distortions, and accumulation effects, posing two critical challenges for deraining. First, collecting clean image pairs for real rainy images is impractical, while existing synthetic datasets still fail to capture the full spectrum of degradation types. Moreover, directly combining datasets may introduce inconsistent distributions due to differences in background quality, res-

olution, and synthesis methods, potentially confusing the model and reducing both performance and generalization. In addition, most existing architectures assume a single degradation type and are specifically tailored to it. In real rainy scenarios, degradations often evolve dynamically during the rain process. Such designs typically suffer noticeable performance declines when applied to unseen rain types, hindering progress from task-specific solutions toward more general intelligence. *This raises a natural question: can we design a unified image deraining framework that adaptively handles diverse rain within a single model?*

To explore this problem, we construct a new rain dataset named Hybrid-Rain10K, a large-scale synthetic dataset built using a Blender-based rendering pipeline. It contains 10,000 paired samples with three representative rain degradations types, (*i.e.*, rain streaks, raindrops and rain accumulation), applied to diverse outdoor backgrounds with fine-grained control over density, angle, transparency, and blur. This dataset provides realistic and diverse rain effects, establishing a comprehensive benchmark for unified image deraining.

Building upon this dataset, we propose a dual mixture-of-experts (MoE) framework. The encoder adopts a Soft MoE to assign collaborative weights across all experts, mitigating early information loss and enhancing complementarity through two-level gating. The decoder introduces a Hard MoE, where top-k routing guided by high-frequency priors enables more effective texture recovery. Extensive experiments show that our model delivers better performance on Hybrid-Rain10K and generalizes strongly across public benchmarks, validating both its effectiveness and robustness.

In this paper, our main contributions are as follows:

- We construct Hybrid-Rain10K, a new large-scale dataset with 10,000 paired samples covering three representative degradations, providing a comprehensive benchmark for unified deraining.
- We propose a Dual Mixture-of-Experts (Dual-MoE) framework that adaptively addresses various degradations types of rain within a single model.
- We demonstrate the effectiveness of our dataset, and show that our proposed method achieves favorable deraining performance against recent ones.

## 2. DATASET CONSTRUCTION

Fig. 2 shows the Hybrid-Rain10K construction pipeline. We use web crawlers to collect background images from open source databases, which are then filtered by an AI system and manually refined to ensure visual quality and scene diversity.

To generate realistic rainy images, we synthesize multiple types of degradations using Blender, an open-source 3D graphics engine. Specifically, there are three types of degradation masks: rain streaks ( $RS$ ), rain drops ( $RD$ ), and rain mist ( $RM$ ). We use Blender’s rain generator  $\tau(\cdot)$  to generate rain streak masks  $M_{RS}$ . Rain mist is modeled as a combination of rain streaks and haze. We use the atmospheric scattering model to synthesize the rain mist mask  $M_{RM}$ . Inspired by UAV-Rain1k [8], the raindrop modeling process is defined as a motion blur process. The process is defined as:

$$M_i = \begin{cases} \tau(d, l, \theta, \omega), & \text{if } (i = RS) \\ \tau(d, l, \theta, \omega) \cdot t + A(1 - t), & \text{if } (i = RM) \\ r_0(1 + \sum_m \cos(m\phi)p_m(\varphi)), & \text{if } (i = RD) \end{cases} \quad (1)$$

where  $d$  denotes the rain streak density,  $l$  the rain streak length,  $\theta$  the angle,  $\omega$  the transparency,  $A$  the atmospheric light,  $t$  the transmittance,  $r_0$  the undeformed radius,  $\phi$  the angle between a given point and the projection of the line of sight onto any plane perpendicular to the direction of fall,  $\cos(m\phi)$  a factor depending on the droplet size  $r_0$ ,  $\varphi$  the angle with the falling direction of the raindrop at that point, and  $p_m(\varphi)$  the function describing the variation of blur morphology with respect to  $\varphi$ . The linear superposition model is used to fuse the degradation mask  $M_i$  with the GT image  $B$ :

$$R_i = B + M_i \quad i \in \{RS, RM, RD\}, \quad (2)$$

where  $R_i$  is the degraded image generated.

Hybrid-Rain10K ultimately contains 10,000 pairs of images with different rain degradation effects. Each ground-truth image is synthetically augmented using  $RS$ ,  $RM$ , or  $RD$  rainfall variants based on scene characteristics, increasing the diversity of the dataset and thus improving generalization capabilities.

## 3. PROPOSED METHOD

### 3.1. Overall Architecture

As illustrated in Fig. 2, the proposed network adopts an asymmetric U-shaped architecture in which distinct experts are wisely assigned to the encoder and decoder stages. Specifically, the input degraded image is first processed by a  $3 \times 3$  convolution layer to extract shallow features. These features are subsequently fed into a four-level encoder, where each level leverages a Soft Mixture-of-Experts (S-MoE) integrated with channel attention to comprehensively capture diverse rain degradation patterns. To further enrich the representation capacity of the decoder, high-frequency guidance is

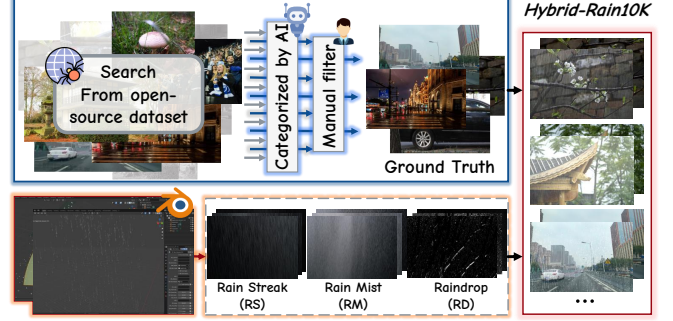


Fig. 1. An overview of the dataset pipeline.

explicitly provided via Sobel-filtered global features, serving as informative cues for subsequent expert selection. The encoded representations are then progressively decoded through a symmetric three-level decoder, within which each block employs a Hard Mixture-of-Experts (H-MoE) with spatial attention to dynamically activate the most appropriate experts according to the encoded features. Finally, a refinement layer is applied to further enhance the restored features before yielding the ultimate clean image, with the entire network optimized under a residual learning paradigm by adding the reconstructed residual to the original input.

### 3.2. Soft Mixture-of-Experts

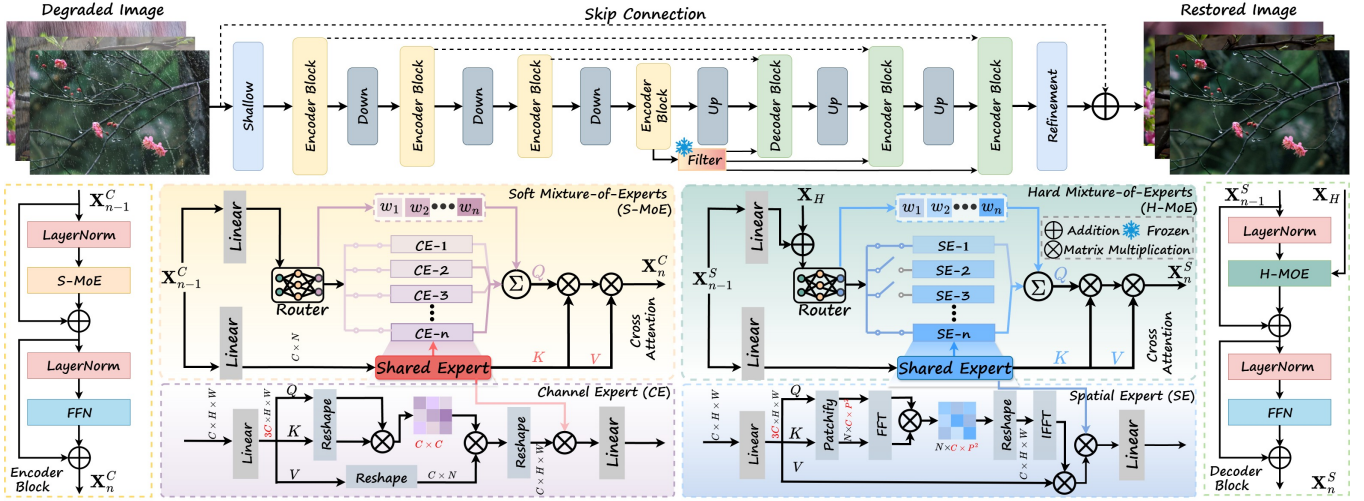
Early Mixture-of-Experts (MoE) architectures often suffer from sparsely activated routing, which risks discarding informative yet low-probability degradation patterns and causing irreversible information loss. To overcome this, we introduce a Soft Mixture-of-Experts (S-MoE) module at the encoder stage. In contrast to hard routing, S-MoE assigns collaborative weights to all experts, enabling exhaustive exploration of diverse rain degradations before any irreversible selection.

Specifically, S-MoE comprises  $n$  channel experts and one shared expert, each tailored to different complexity regimes of rain degradation patterns. To enhance representational complementarity, we employ a two-level gating mechanism to orchestrate the interaction between the specialized channel experts and the shared expert. Specifically, routed tokens are first projected into expert-specific embeddings, followed by a modulation process that fuses specialized and shared representations, and finally integrated via a cross-attention operation. Mathematically, the processing of  $i$ -th expert handling input features  $\mathcal{X}$  through a channel expert can be defined as:

$$\mathcal{Y}_{CE}^i = \tilde{\mathcal{X}}_{CE}^i \otimes \mathcal{S}_c(\mathcal{X}), \mathcal{S}_c(\mathcal{X}) = \text{MHCA}(\mathcal{X}), \quad (3)$$

$$\tilde{\mathcal{X}}_{CE}^i = \mathcal{CE}^i(\mathcal{X}_{CE}^i) = \text{MHCA}_{r_i}(\mathcal{L}_C^i(\mathcal{X})), \quad (4)$$

where  $\mathcal{Y}_{CE}^i$  denotes the  $i$ -th modulated expert features,  $\tilde{\mathcal{X}}_{CE}^i$  is the input of  $i$ -th expert,  $\otimes$  is the element-wise multiplication and  $i$  is the index of the current channel expert  $\mathcal{CE}^i$  with the multi-head channel attention (MHCA) [9] and embedding



**Fig. 2.** Overall architecture of UD<sup>2</sup>MoE for unified image deraining, incorporating a dual mixture-of-experts (MoE) model.

dimensionality  $r_i$ .  $\mathcal{S}_e$  denotes the shared channel expert and  $\mathcal{L}$  indicates the linear layers for the projection of input features.

Distinct from token routing, our design operates at the image level to avoid fragmented representations:

$$\mathcal{G}_{en} = \sigma(\mathcal{W}(\varphi(\mathcal{X} + \epsilon))), \quad (5)$$

where  $\mathcal{G}_{en}$  denotes the routing function,  $\mathcal{W}$  is learnable projection matrices,  $\varphi$  denotes global average pooling and  $\epsilon \sim \mathcal{N}(0, \sigma^2)$  is training noise for load balancing.  $\sigma$  denotes the softmax operator. The final output  $\mathcal{Y}_{CE}$  combines all channel experts contributions through weighted aggregation.

$$\mathcal{Y}_{CE} = \sum_{i=1}^N \mathcal{G}_{en}^i \otimes \mathcal{Y}_{CE}^i, \quad (6)$$

where  $\otimes$  is the final output. This design ensures both diversity and coherence in the extracted features, thereby enhancing the ability of the encoder to model diverse rain patterns.

### 3.3. Hard Mixture-of-Experts

Although S-MoE employs soft routing to comprehensively capture diverse degradation patterns, the decoder faces distinct challenges that require efficient and accurate feature reconstruction. To this end, we propose a Hard Mixture-of-Experts (H-MoE) with a hard routing strategy that activates only the  $k$  most relevant experts, significantly reducing computational overhead while preserving essential features. Frequency guidance is incorporated during expert selection to prioritize texture-sensitive experts, enhancing high-frequency detail recovery. To further improve the optimized efficiency, we leverage an FFT-based approximation [10] for feature reconstruction. Akin to the framework of S-MoE, H-MoE also comprises  $n$  spatial experts of varying complexity and a single shared expert, and adopts a two-level gating mechanism to modulate the final output features. The processes of the  $i$ -th spatial expert handling input features  $\mathcal{X}$  is defined as:

$$\mathcal{Y}_{SE}^i = \tilde{\mathcal{X}}_{SE}^i \odot \mathcal{S}_s(\mathcal{X}), \mathcal{S}_s(\mathcal{X}) = \text{FFT-SA}(\mathcal{X}), \quad (7)$$

$$\tilde{\mathcal{X}}_{SE}^i = \mathcal{S}\mathcal{E}^i(\mathcal{X}_{SE}^i) = \text{FFT-SA}_{p_i}(\mathcal{L}_S^i(\mathcal{X}) + \mathcal{X}_H), \quad (8)$$

where  $\mathcal{Y}_{SE}^i$  denotes the output of  $i$ -th spatial expert,  $\tilde{\mathcal{X}}_{SE}^i$  is the input of  $i$ -th expert,  $i$  is the index of the current spatial expert  $\mathcal{S}\mathcal{E}^i$  with the FFT-SA and patch size  $p_i$ .  $\mathcal{S}_s$  denotes the shared spatial expert and  $\mathcal{X}_H$  is the high-frequency feature.

Then, H-MoE adopts a routing function  $\mathcal{G}_{de}$  that orchestrates the allocation of input to the most suitable expert by the top- $k$  strategy  $\mathcal{T}_k$ . This processing can be defined as:

$$\mathcal{G}_{de} = \mathcal{T}_k(\sigma(\mathcal{W}(\varphi(\mathcal{X} + \epsilon) + \mathcal{X}_H))), \quad (9)$$

$$\mathcal{Y}_{SE} = \sum_{i=1}^N \mathcal{G}_{de}^i \cdot \mathcal{Y}_{SE}^i. \quad (10)$$

Following the above design, H-MoE enhances the ability to capture structural details while balancing computational efficiency with both global and local detail reconstruction.

## 4. EXPERIMENTS

### 4.1. Implementation Details

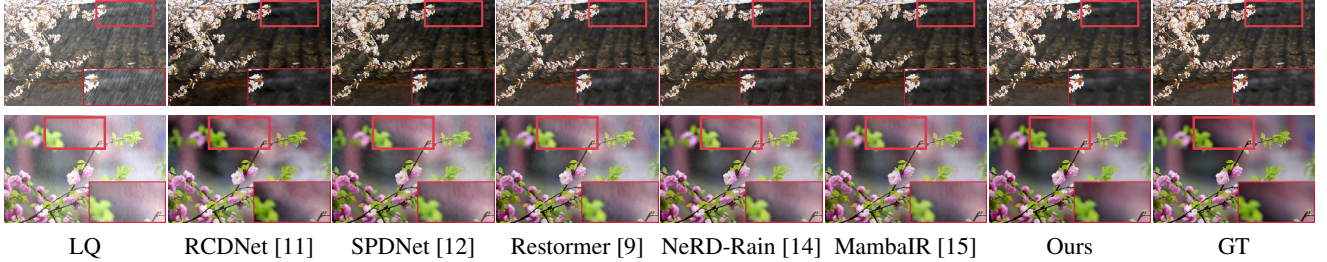
The UD<sup>2</sup>MoE framework is end-to-end trainable and adopts a symmetric encoder-decoder architecture. The encoder has four stages, with [4, 6, 6, 8] transformer modules from top to bottom. The decoder includes three levels, containing [2, 4, 4] transformer blocks. In each MoE layer,  $n = 4$  nested experts are used. Epoch is set to 120 with a batch size of 32 and a patch size of  $120 \times 128$ . The Adam optimizer is used with an initial learning rate of  $2 \times 10^{-4}$  and cosine decay. Training is conducted on a single NVIDIA GeForce RTX 4090 GPU.

### 4.2. Quantitative Comparisons

To ensure fair comparison, all compared algorithms are re-trained on the proposed dataset. Table 1 compares the results of UD<sup>2</sup>MoE with several existing state-of-the-art methods on the Hybrid-Rain10K dataset. The deraining performance is evaluated using three benchmark metrics: PSNR [16], SSIM [17], and LPIPS [18]. Our UD<sup>2</sup>MoE achieves

**Table 1.** Quantitative evaluations on the Hybrid-Rain10K dataset. The best and second-best values are **bold** and underlined.

| Model/Metrics              | Venue      | RS              |                 |                    | RD              |                 |                    | RM              |                 |                    | Average         |                 |                    |
|----------------------------|------------|-----------------|-----------------|--------------------|-----------------|-----------------|--------------------|-----------------|-----------------|--------------------|-----------------|-----------------|--------------------|
|                            |            | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ |
| RCDNet [11]                | CVPR 2020  | 26.5615         | 0.8324          | 0.2374             | 28.5235         | 0.8199          | 0.3015             | 26.6063         | 0.8566          | 0.2638             | 23.4135         | 0.7047          | 0.2803             |
| SPDNet [12]                | ICCV 2021  | 24.3025         | 0.8423          | 0.2748             | 28.7572         | 0.8260          | 0.3108             | 27.0856         | 0.8426          | 0.2295             | 23.0310         | 0.7110          | 0.2899             |
| Restormer [9]              | CVPR 2022  | 31.0060         | 0.9175          | 0.1153             | 30.2229         | 0.8507          | 0.2604             | 24.9039         | 0.8739          | 0.2032             | 28.6688         | 0.8756          | 0.2020             |
| IDT [13]                   | TPAMI 2022 | 32.0291         | <u>0.9306</u>   | <u>0.0756</u>      | 31.8091         | 0.8782          | 0.1812             | 29.7008         | <u>0.9261</u>   | <u>0.1087</u>      | 31.0661         | 0.9022          | 0.1393             |
| DRSformer [1]              | CVPR 2023  | 29.0868         | 0.8758          | 0.1827             | 30.0675         | 0.8719          | 0.2227             | 25.5606         | 0.8701          | 0.2343             | 28.3669         | 0.8726          | 0.2192             |
| MSDT [2]                   | AAAI 2024  | 30.1303         | 0.8852          | 0.1603             | <u>32.9311</u>  | <u>0.9017</u>   | <u>0.1456</u>      | 28.7865         | 0.8930          | 0.1878             | 31.7961         | <u>0.9107</u>   | <u>0.1309</u>      |
| NeRD-Rain [14]             | CVPR 2024  | <u>32.3051</u>  | 0.9250          | 0.0965             | 32.8018         | <u>0.9017</u>   | 0.1497             | <u>30.2876</u>  | 0.9155          | 0.1362             | <u>31.8373</u>  | 0.9100          | 0.1372             |
| MambaIR [15]               | ECCV 2024  | 28.5479         | 0.8583          | 0.1960             | 30.7564         | 0.8614          | 0.2394             | 24.6125         | 0.8387          | 0.2705             | 28.7242         | 0.8664          | 0.2174             |
| UD <sup>2</sup> MoE (Ours) | -          | <b>32.5753</b>  | <b>0.9317</b>   | <b>0.0715</b>      | <b>35.3646</b>  | <b>0.9419</b>   | <b>0.0715</b>      | <b>30.3668</b>  | <b>0.9354</b>   | <b>0.0943</b>      | <b>33.1859</b>  | <b>0.9381</b>   | <b>0.0796</b>      |

**Fig. 3.** Qualitative comparison for different networks on our proposed Hybrid-Rain10K dataset. Zoom in for a better view.**Table 2.** Ablation on model variants.

| Methods   | Encoder      |              | Decoder      |              | Average         |                 |                    |
|-----------|--------------|--------------|--------------|--------------|-----------------|-----------------|--------------------|
|           | H-MoE        | S-MoE        | H-MoE        | S-MoE        | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ |
| Baseline  | $\times$     | $\times$     | $\times$     | $\times$     | 30.1274         | 0.8943          | 0.1598             |
| Variant 1 | $\checkmark$ | $\times$     | $\checkmark$ | $\times$     | 32.9951         | 0.9371          | 0.0812             |
| Variant 2 | $\checkmark$ | $\times$     | $\times$     | $\checkmark$ | 32.8695         | 0.9363          | 0.0811             |
| Variant 3 | $\times$     | $\checkmark$ | $\times$     | $\checkmark$ | 32.6975         | 0.9362          | 0.0839             |
| Ours      | $\times$     | $\checkmark$ | $\checkmark$ | $\times$     | <b>33.1859</b>  | <b>0.9381</b>   | <b>0.0796</b>      |

state-of-the-art performance on all three subsets and on average. On the severely degraded RD subset, our method outperforms the suboptimal MSDT [2] by approximately 2.4335dB, 0.0402, and 0.0741 in PSNR, SSIM, and LPIPS, respectively. Compared to NeRD-Rain [14], UD<sup>2</sup>MoE improves the average PSNR by 1.3486 dB. Compared to MSDT [2], it outperforms the average SSIM by 0.0274 and the average LPIPS by 0.0513. These results demonstrate our method’s robustness.

### 4.3. Qualitative Comparisons

Fig. 3 shows the visual performance of our method on the Hybrid-Rain10K dataset. The first row of the figure shows that our method achieves superior visual results. While other methods still retain a large number of rain streaks, our method removes them and restores a clear background image. The results in the second row demonstrate that even with more complex rain mist degradation, our method can still accurately restore the background image and preserve the original color and detail. However, other methods (e.g. SPDNet [12] and MambaIR [15]) still retain a large amount of rain mist, significantly degrading the visual quality of the image. The qualitative results intuitively show that our method can not only remove rain streaks and restore a clean background image, but also preserve the color and detail information of the image.

### 4.4. Ablation Studies

To validate the effectiveness and rationality of the Dual-MoE framework adopted in our approach, we provide ablation results in Table 2. The baseline model achieves the lowest performance due to the removal of all MoE modules, demonstrating that the simple backbone network is insufficient to model complex and diverse rain degradation. Introducing H-MoE in both the encoder and decoder significantly improves performance, with PSNR and SSIM increasing by 3.0585 dB and 0.0438, respectively, while reducing LPIPS by 0.0802. This demonstrates its ability to comprehensively capture heterogeneous degradation patterns and prevent the loss of early information features. Using S-MoE in both the encoder and decoder results in decreased performance, suggesting that the soft routing mechanism may not fully model degradation information. Our model uses S-MoE in the encoder to assign weights to all experts for exhaustive exploration, and H-MoE in the decoder to activate relevant experts and reduce computational overhead. These results show S-MoE and H-MoE are complementary, with our model achieving top performance.

## 5. CONCLUSION

In this work, we introduce Hybrid-Rain10K, a large-scale dataset with three representative rain degradations for unified image deraining. We further propose a Dual-MoE framework combining a Soft-MoE encoder and a Hard-MoE decoder guided by high-frequency priors. Experiments confirm its superior performance and strong cross-dataset generalization. Future work will explore dynamic rain scenarios and unsupervised learning for real-world adaptation.



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